

JEE MAIN REVISION MASTER: GENERAL ORGANIC CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Acidity, Basicity & Aromaticity (Huckel's Rule)	5 PYQs	Physical-Chemical Properties
Moderate	Reaction Intermediates & Electronic Effects	4 PYQs	Core Principles

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Acidity, Basicity & Aromaticity (Huckel's Rule)	$\text{Aromatic: } (4n+2)\pi \text{ electrons}, \quad \text{Anti-aromatic: } 4n\pi \text{ electrons}$
Reaction Intermediates & Electronic Effects	$\text{Carbocation Stability: } 3^\circ > 2^\circ > 1^\circ > \text{CH}_3^+, \quad \text{Carbanion Stability: } \text{CH}_3^- > 1^\circ > 2^\circ > 3^\circ$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Acidity, Basicity & Aromaticity (Huckel's Rule) (Moderate Yield — 5 PYQs)

Core Principle: Hückel's Rule: Cyclic, planar, completely conjugated systems with $(4n+2)\pi$ electrons are aromatic (2, 6, 10, 14π). Basicity of amines: Aliphatic amines are more basic than aniline because aniline's lone pair is delocalized into the benzene ring by resonance.

■■ **Caution / JEE Pitfall:** Cyclooctatetraene (8π) is non-planar (tub-shaped), hence it is non-aromatic, NOT anti-aromatic!

■ **Speed Trick:** In gas phase, amine basicity follows strictly $+I: 3^\circ > 2^\circ > 1^\circ > \text{NH}_3$.

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "Which of the following is not an aromatic compound..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "Among the following. The set which represents aromatic species is..." [Ans: N/A]
- [2021 Shift] Q#4 (MCQ): "Arrange benzene, *n*-hexane and ethyne in decreasing order of their acidic behaviour...." [Ans: N/A]

3.2 Reaction Intermediates & Electronic Effects (Moderate Yield — 4 PYQs)

Core Principle: Heterolytic fission creates ions (carbocations, carbanions); homolytic fission creates neutral free radicals. Stability of carbocations: $3^\circ > 2^\circ > 1^\circ > \text{CH}_3^+$ (governed by hyperconjugation and +I). Stability of carbanions: $\text{CH}_3^- > 1^\circ > 2^\circ > 3^\circ$.

■ **Caution / JEE Pitfall:** In homolytic cleavage, each atom takes one bonding electron; in heterolytic cleavage, the more electronegative atom takes both electrons.

■ **Speed Trick:** Resonance stabilization always dominates over inductive effect, EXCEPT for halogens on benzene rings where -I dominates.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The intermediates in heteropolar reactions are..." [Ans: N/A]
- [2021 Shift] Q#4 (MCQ): "Arrange benzene, n -hexane and ethyne in decreasing order of their acidic behaviour..." [Ans: N/A]
- [2019 Shift] Q#2 (MCQ): "Resonance effect is not observed in..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The intermediates in heteropolar reactions are

- (A) Free radicals only
- (B) Cations only
- (C) Anions only
- (D) Both anions and cations

Correct Answer: None

Solution Analysis: In heteropolar reactions, the bond breaks or forms in such a way that the electron pair goes to one atom. This gives charged species. For example, in heterolytic bond fission: $\text{A-B} \rightarrow \text{A}^+ + \text{B}^-$. So the intermediates formed can be positive ions, called cations, or negative ions, called anions. Free radicals are formed in homolytic bond fission, not in heteropolar reactions. Hence, the correct option is: Option D: Both anions and cations

Benchmark Problem 2: KCET 2023 (Q#3)

Among the following. The set which represents aromatic species is

- (A) I, II and III
- (B) III, IV and V
- (C) II and III
- (D) I, II and IV

Correct Answer: None

Solution Analysis: Compounds I, II and IV have continuous delocalisation of π -electrons and have $(4n+2\pi)$ electrons. On the other hand compound III does not show conjugation and compound V has $4n\pi$ electrons. Thus, compounds given in only I, II and IV are aromatic.